MODULE 20 · DOMAIN 4 - EVALUATION, TESTING AND OPTIMISATION

Diagnosing Failures

A method for any failure, symptom-to-layer reasoning, telling hallucination from a retrieval miss or a prompt problem, and fixing at the right layer.

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The plumber's first question is not what is wrong. It is **what changed?**

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A diagnostic method

01
What changed?
Deploys, versions, index, config, traffic.

02
Reproduce
The smallest failing case.

03
Isolate the layer
Retrieval, prompt, model, tool, config.

04
Compare
Against the last known good.

05
Fix one thing
At the layer that failed.

06
Lock it in
Add the case to the eval set.

ON THE EXAM

Diagnose before you change anything. Every fix without a diagnosis is a guess that adds a variable.

From symptom to layer

SYMPTOM	LOOK FIRST AT
Documents refreshed; model and latency unchanged	Retrieval and the index
Behaviour shifted after a model upgrade	Model mismatch: replay on the previous version
Broke after a maintenance window or deploy	Configuration: diff against last known good
Answer contradicts sources that were retrieved	Grounding instructions
Fails only for some inputs or groups	Coverage, data and the prompt for that slice
Slow drift, nothing deployed	The input mix: monitoring and samples

ON THE EXAM

The timing of a failure is evidence. Line it up with the change log before forming a theory.

Failures that look alike

FAILURE	IN THE TRACE YOU SEE	CONFIRM BY
Retrieval miss	The right passage was never in context	Checking the retrieved chunks
Hallucination	A claim no supplied passage supports	Matching each claim to its source
Prompt failure	Context fine; instructions unclear or in conflict	Reading the prompt against the behaviour
Model mismatch	Same inputs, different behaviour by version	Replaying on the previous version
Tool or config fault	Bad data or settings reaching the model	Inspecting raw tool and model output

ON THE EXAM

Same symptom, a wrong answer, five different causes. The trace tells them apart.

Read the raw evidence

- **The transcript:** every step, tool call and result, not just the final answer.
- **Raw output before parsing:** parsers hide truncation and malformed fields.
- **The stop reason:** max_tokens means the answer was cut off, not finished.
- **Versions and request IDs** for the failing call and the last good one.
- **Never diagnose from a summary** written by the system being diagnosed.

> ON THE EXAM
>
> Anthropic's advice for evals applies here too: read the transcripts.

Anthropic's hallucination controls, October 2026

CONTROL	HOW IT HELPS
Allow I don't know	Removes pressure to invent an answer
Quotes first, for long documents	Grounds the task in the actual text
Cite, or retract	Each claim needs a supporting quote, or it goes
Supplied documents only	Restricts answers to the provided sources
Best-of-N comparison	Inconsistency across runs flags invention
Review summarised thinking	Shows faulty reasoning behind a wrong answer

ON THE EXAM

Anthropic: these reduce hallucinations but do not eliminate them. Validate critical information.

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Fix at the layer that failed

LAYER	FIXES THAT BELONG THERE
Retrieval	Re-index, re-chunk, hybrid search, metadata filters
Grounding	Supplied-sources-only instructions, quotes, citations
Prompt	Resolve conflicts, clarify, add examples
Model	Re-tune prompts for the version, or roll back while you do
Tool or config	Fix the data, restore the last known good settings

THE TIEBREAKER

Lower the temperature is never the fix. Wrong layer, and current Opus models reject it anyway.

> Two days after Tallyhawk rewrote 1,200 help articles for a major release, its support agent began giving confident, outdated answers.
>
> The model version and latency are unchanged.

What should Corvane investigate first?

A Replay the failing questions on the previous model version

B Rewrite the grounding rules in the system prompt

C Raise the effort level so the agent reasons more carefully

D Whether the index and retrieved chunks reflect the rewritten articles

◆ PAUSE AND DECIDE

What should Corvane investigate first?

A Replay the failing questions on the previous model version

The model did not change.

B Rewrite the grounding rules in the system prompt

It is faithfully grounding in what it is given.

C Raise the effort level so the agent reasons more carefully

Better reasoning over stale passages.

D Whether the index and retrieved chunks reflect the rewritten articles

Documents changed; nothing else did. Check retrieval.

THE SIGNAL

Documents refreshed, model and latency unchanged, answers wrong: retrieval and the index first.

> Mereside migrated its complaints summaries to a newer model version, because the old one is retiring. The prompt did not change.
>
> About one summary in ten now omits the remedy the customer asked for.

What is the best first diagnostic step?

A. Rewrite the prompt immediately to emphasise remedies

B. Replay the same complaints on the previous version and compare outputs side by side

C. Lower the temperature to make summaries more consistent

D. Move the summaries to the heaviest model tier

◆ PAUSE AND DECIDE

What is the best first diagnostic step?

A Rewrite the prompt immediately to emphasise remedies

A fix before a diagnosis. It may be right, but prove it.

B Replay the same complaints on the previous version and compare outputs side by side

Isolates the version as the cause, on identical inputs.

C Lower the temperature to make summaries more consistent

Wrong layer, and rejected by current Opus models.

D Move the summaries to the heaviest model tier

Another variable, before the first is understood.

THE SIGNAL

Behaviour shifted after a model change: replay on the previous version to confirm model mismatch.

> After a weekend maintenance window, Quillon's discharge-summary pipeline returns empty medication lists for some long admissions.
>
> No model or prompt change was announced.

What should Corvane do first?

A. Diff the configuration against the last known good, and inspect the raw model output before parsing

B. Add more examples of medication lists to the prompt

C. Move the pipeline to a more capable model tier

D. Ask clinicians to complete the lists by hand until it resolves

◆ PAUSE AND DECIDE

What should Corvane do first?

A Diff the configuration against the last known good, and inspect the raw model output before parsing

Maintenance changed something; the raw output shows what.

B Add more examples of medication lists to the prompt

The prompt worked last week.

C Move the pipeline to a more capable model tier

A new variable, not a diagnosis.

D Ask clinicians to complete the lists by hand until it resolves

A workaround that hides the fault.

THE SIGNAL

Broke after maintenance: diff the configuration, and read the raw output before the parser.

> Mereside's agent assist told an adviser that early repayment charges never apply after five years. The trace shows the correct policy passage was retrieved.
>
> That passage lists exceptions where the charge does apply. The answer contradicts it.

Which fix fits the failure?

A Rebuild the index for the lending policy

B Increase the chunk size so more policy text is retrieved

C Strengthen grounding: answer only from the supplied passages, quote them, and say when they do not cover the question

D Switch to a different embedding model

◆ PAUSE AND DECIDE

Which fix fits the failure?

A Rebuild the index for the lending policy

Retrieval worked. The passage was there.

B Increase the chunk size so more policy text is retrieved

More text the answer may still ignore.

C Strengthen grounding: answer only from the supplied passages, quote them, and say when they do not cover the question

The answer overrode its source. Fix the grounding.

D Switch to a different embedding model

Retrieval is not the failing layer.

THE SIGNAL

Right passage retrieved, answer contradicts it: a grounding failure, not a retrieval one.

> A Quillon clinician found a discharge summary citing a lab result that is not in the patient's record.
>
> The trace shows the record sections supplied to Claude contained no such result. Nobody had sampled summaries since launch.

How should Corvane classify and address this?

A. A retrieval miss: rebuild the index and re-run

B. A prompt conflict: rewrite the system prompt from scratch

C. Model mismatch: roll back to the previous model version

D. A hallucination: require citations to supplied passages, allow not in the record, and sample summaries for review

◆ PAUSE AND DECIDE

How should Corvane classify and address this?

A. A retrieval miss: rebuild the index and re-run
The result does not exist. Nothing was missed.

B. A prompt conflict: rewrite the system prompt from scratch
No evidence of conflicting instructions.

C. Model mismatch: roll back to the previous model version
Nothing changed version. No comparison suggests it.

D. A hallucination: require citations to supplied passages, allow not in the record, and sample summaries for review
An unsupported claim: ground it, permit gaps, and watch for it.

THE SIGNAL

A claim no supplied source supports is a hallucination. Cite, permit I don't know, and sample.

> Tallyhawk's support quality scores dropped sharply on Wednesday. The team proposes rewriting the whole system prompt.
>
> Corvane has access to the change log, traces and the evaluation set.

What should Corvane do first?

A. Find what changed around Wednesday: deploys, model, prompt, index, configuration and traffic mix

B. Rewrite the system prompt, as the team proposes

C. Switch to a more capable model tier

D. Collect another month of data before acting

◆ PAUSE AND DECIDE

What should Corvane do first?

A Find what changed around Wednesday: deploys, model, prompt, index, configuration and traffic mix

A sharp drop on a date has a cause on that date.

B Rewrite the system prompt, as the team proposes

A big change before a diagnosis adds variables.

C Switch to a more capable model tier

Another untested change.

D Collect another month of data before acting

Customers suffer while the cause sits in the log.

THE SIGNAL

A sharp drop on a known date: start with the change log, not a rewrite.

> Quillon's portal assistant gives some patients appointment times that are hours off. It reads times from the scheduling tool.
>
> The trace shows the tool returns times in UTC, unlabelled, and Claude presents them as local time.

Which layer failed, and what is the fix?

A. A hallucination: add citation requirements to the prompt

B. A retrieval miss: re-index the appointment data

C. The tool integration: return local times, or label the time zone, and add the case to the eval set

D. The model tier: move to a model that infers time zones better

◆ PAUSE AND DECIDE

Which layer failed, and what is the fix?

A A hallucination: add citation requirements to the prompt

Claude repeated the tool's value. It invented nothing.

B A retrieval miss: re-index the appointment data

The right record arrived. Its format misled.

C The tool integration: return local times, or label the time zone, and add the case to the eval set

Fix the data at its source, and lock the case in.

D The model tier: move to a model that infers time zones better

Inference cannot recover information the tool never gave.

THE SIGNAL

Bad or ambiguous tool data reaching the model: fix the tool's output, not the model.

> Tallyhawk's billing agent started misapplying the loyalty discount yesterday. An engineer has a fix ready.
>
> Corvane asks which steps must happen before the fix is applied.

Which two steps come before applying the fix? Choose two.

- A Promote the fix to all customers
- B Close the incident in the tracker
- C Rewrite the agent's prompt from scratch
- D Reproduce the failure with a minimal case
- E Compare the failing configuration with the last known good

◆ PAUSE AND DECIDE

Which two steps come before applying the fix? Choose two.

A Promote the fix to all customers

After the fix is proven, not before.

B Close the incident in the tracker

Last, once the case is in the eval set.

C Rewrite the agent's prompt from scratch

A new variable, not a step in diagnosis.

D Reproduce the failure with a minimal case

Proves the fault, and later proves the fix.

E Compare the failing configuration with the last known good

Shows what changed, so the fix targets it.

THE SIGNAL

Reproduce and compare before you fix. A fix you cannot test against a reproduction is a hope.

In five lines

- **Ask what changed,** reproduce minimally, isolate the layer, compare with last known good.
- **Timing points to the layer:** documents, model upgrade, maintenance, or drift.
- **The trace separates look-alikes:** miss, hallucination, prompt, model, tool.
- **Read raw evidence:** transcripts, unparsed output, the stop reason.
- **Fix one thing at the failing layer,** then add the case to the eval set.

COMING UP

Next: optimising cost and latency, and monitoring in production.